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Date: - 14/09/2026 Programme: - B. TECH (CSE)

Assignments: - Write a menu-drive C programme that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:
Print the first 10 multiples of each integer from m to n .
Print the addition table of each integer from m to n , using addends From 1 to 10. Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.
Exit the programme.
Display an appropriate error message if the user enters an invalid menu choice. Continue displaying the menu and performing the selected operation until the user chooses the Exit option.
Note: - Prepare the algorithm and test-case table. Then write, compile and execute the C programme, and test it using the prepared test cases.

Part 1: - Thorough understanding of the problem statement

Input: -

m (Lower bound integer, Variable type: *int*, Unit: unitless integer)
 n (Upper bound integer, Variable type: *int*, Unit: unitless integer)
choice (Menu option selection, Variable type: *int*, Unit: unitless integer)

Output: -

Multiplication, Addition, and Subtraction table values (Variable type: *int*)
Division table values (Variable type: *float* or *double*, formatted to 2 decimal places)

Formula: -

Multiples: - $result = i * j$;

Addition: - $result = i + j$;

Subtraction: - $result = i - j$;

Division: - $result = (float)i / j$; (where i iterates from m to n , and j iterates from 1 to 10)

Part 2: - Writing, compiling and evaluating the programme

Step 1: - Start

Step 2: - Declare integer variables m , n , *choice*, i and j .

- Step 3:** - Prompt the user to enter two positive integers, m and n .
- Step 4:** - Validate the input to ensure $m > 0$, $n > 0$, and $m \leq n$. If invalid, display an error and terminate or prompt again.
- Step 5:** - Begin a *do-while* loop to display the menu.
- Step 6:** - Display the menu options
- 1: Multiples
 - 2: Addition
 - 3: Subtraction
 - 4: Division
 - 5: Exit
- and read *choice*.
- Step 7:** - Initiate a *switch-case* block based on *choice*.
- Step 8(Case 1):** - Loop i from m to n . Inside, loop j from 1 to 10. Print $i * j$.
- Step 9(Case 2):** - Loop i from m to n . Inside, loop j from 1 to 10. Print $i + j$.
- Step 10(Case 3):** - Loop i from m to n . Inside, loop j from 1 to 10. Print $i - j$.
- Step 11(Case 4):** - Loop i from m to n . Inside, loop j from 1 to 10. Print $(float)i / j$ formatted to $\%.2f$.
- Step 12(Case 5):** - Print an exit message and break the loop.
- Step 13(Case 6):** - Print an "invalid choice" error message.
- Step 14:** - Stop

Part 3: - Writing and compiling the C programme

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~
Session Actions Edit View Help
GNU nano 9.0 math tables.c *
#include <stdio.h>

int main() {
    int m, n, choice, i, j;

    // Input and validation
    printf("Enter two positive integers m and n (where m <= n):\n");
    printf("Enter m: ");
    scanf("%d", &m);
    printf("Enter n: ");
    scanf("%d", &n);

    if (m <= 0 || n <= 0 || m > n) {
        printf("Error: Both numbers must be positive integers and m must be less than or equal to n.\n");
        return 1;
    }

    // Menu loop
    do {
        printf("\n--- Mathematical Tables Menu ---\n");
        printf("1. Print the first 10 multiples\n");
        printf("2. Print the addition table (addends 1 to 10)\n");
        printf("3. Print the subtraction table (subtrahends 1 to 10)\n");
        printf("4. Print the division table (divisors 1 to 10)\n");
        printf("5. Exit\n");
        printf("Enter your choice (1-5): ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (i = m; i <= n; i++) {
                    printf("\nMultiples of %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d x %d = %d\n", i, j, i * j);
                    }
                }
                break;

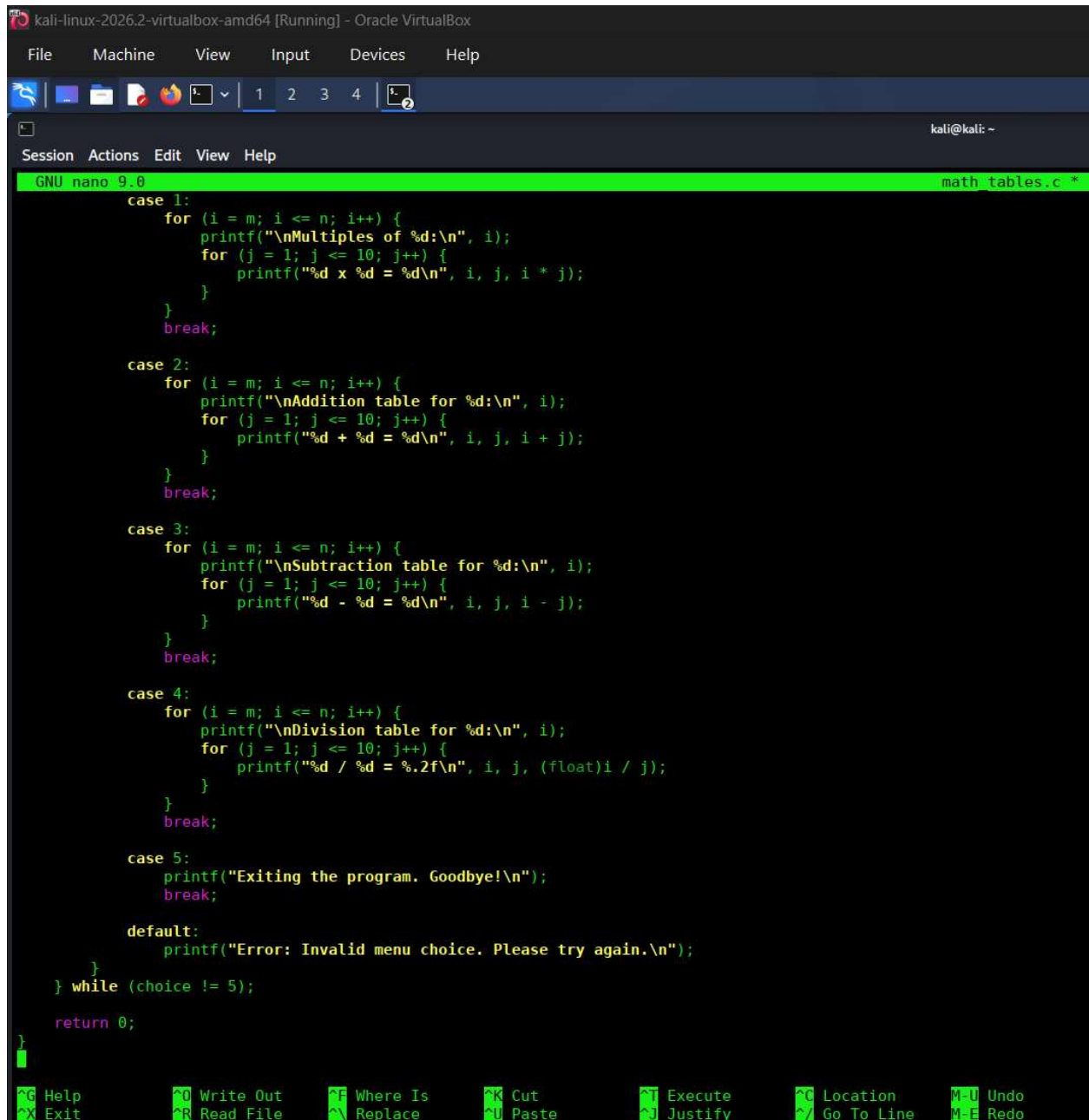
            case 2:
                for (i = m; i <= n; i++) {
                    printf("\nAddition table for %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d + %d = %d\n", i, j, i + j);
                    }
                }
                break;

            case 3:
                for (i = m; i <= n; i++) {
                    printf("\nSubtraction table for %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d - %d = %d\n", i, j, i - j);
                    }
                }
                break;

            case 4:
                for (i = m; i <= n; i++) {
                    printf("\nDivision table for %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d / %d = %d\n", i, j, i / j);
                    }
                }
                break;

            case 5:
                return 0;
        }
    } while (choice != 5);
}
```

Help Write Out Where Is Cut Execute Location M-U Undo
Exit Read File Replace Paste Justify Go To Line W-E Redo



The screenshot shows a Kali Linux virtual machine running Oracle VM VirtualBox. The terminal window displays the GNU nano 9.0 text editor editing a file named `math_tables.c`. The code is a C program that generates multiplication, addition, subtraction, and division tables for a given number `n` (ranging from 1 to 10). It uses a `switch` statement with `case` labels for each operation. The program also includes a `default` case for invalid menu choices and a `while` loop to keep the program running until the user chooses to exit (option 5). The bottom of the terminal shows a status bar with various keyboard shortcuts for nano, such as `Ctrl+G` for Help, `Ctrl+O` for Write Out, etc.

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
kali@kali: ~
Session Actions Edit View Help
GNU nano 9.0 math_tables.c *
case 1:
    for (i = m; i <= n; i++) {
        printf("\nMultiples of %d:\n", i);
        for (j = 1; j <= 10; j++) {
            printf("%d x %d = %d\n", i, j, i * j);
        }
    }
    break;

case 2:
    for (i = m; i <= n; i++) {
        printf("\nAddition table for %d:\n", i);
        for (j = 1; j <= 10; j++) {
            printf("%d + %d = %d\n", i, j, i + j);
        }
    }
    break;

case 3:
    for (i = m; i <= n; i++) {
        printf("\nSubtraction table for %d:\n", i);
        for (j = 1; j <= 10; j++) {
            printf("%d - %d = %d\n", i, j, i - j);
        }
    }
    break;

case 4:
    for (i = m; i <= n; i++) {
        printf("\nDivision table for %d:\n", i);
        for (j = 1; j <= 10; j++) {
            printf("%d / %d = %.2f\n", i, j, (float)i / j);
        }
    }
    break;

case 5:
    printf("Exiting the program. Goodbye!\n");
    break;

default:
    printf("Error: Invalid menu choice. Please try again.\n");
} while (choice != 5);
return 0;
}
Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo
```

Part 4: - Running and executing the algorithm on a platform (Kali Linux VM)

- Step 1: - Open the terminal
- Step 2: - Create the source code
- Step 3: - Write and Save Code
- Step 4: - Compile the programme
- Step 5: - Run the Binary Executable Programme

```
(kali㉿kali)-[~]  
$ nano math_tables.c
```

```
(kali㉿kali)-[~]  
$ gcc math_tables.c -o math_tables
```

```
(kali㉿kali)-[~]  
$ ./math_tables
```

Enter two positive integers m and n (where m <= n):

Enter m: 2

Enter n: 3

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 1

Multiples of 2:

2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiples of 3:

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 2

Addition table for 2:

$2 + 1 = 3$
 $2 + 2 = 4$
 $2 + 3 = 5$
 $2 + 4 = 6$
 $2 + 5 = 7$
 $2 + 6 = 8$
 $2 + 7 = 9$
 $2 + 8 = 10$
 $2 + 9 = 11$
 $2 + 10 = 12$

Addition table for 3:

$3 + 1 = 4$
 $3 + 2 = 5$
 $3 + 3 = 6$
 $3 + 4 = 7$
 $3 + 5 = 8$
 $3 + 6 = 9$
 $3 + 7 = 10$
 $3 + 8 = 11$
 $3 + 9 = 12$
 $3 + 10 = 13$

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 2

Addition table for 2:

$2 + 1 = 3$
 $2 + 2 = 4$
 $2 + 3 = 5$
 $2 + 4 = 6$
 $2 + 5 = 7$
 $2 + 6 = 8$
 $2 + 7 = 9$
 $2 + 8 = 10$
 $2 + 9 = 11$
 $2 + 10 = 12$

Addition table for 3:

$3 + 1 = 4$
 $3 + 2 = 5$
 $3 + 3 = 6$
 $3 + 4 = 7$
 $3 + 5 = 8$
 $3 + 6 = 9$

Addition table for 3:

$3 + 1 = 4$
 $3 + 2 = 5$
 $3 + 3 = 6$
 $3 + 4 = 7$
 $3 + 5 = 8$
 $3 + 6 = 9$
 $3 + 7 = 10$
 $3 + 8 = 11$
 $3 + 9 = 12$
 $3 + 10 = 13$

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 1

Multiples of 2:

$2 \times 1 = 2$
 $2 \times 2 = 4$
 $2 \times 3 = 6$
 $2 \times 4 = 8$
 $2 \times 5 = 10$
 $2 \times 6 = 12$
 $2 \times 7 = 14$
 $2 \times 8 = 16$
 $2 \times 9 = 18$
 $2 \times 10 = 20$

Multiples of 3:

$3 \times 1 = 3$
 $3 \times 2 = 6$
 $3 \times 3 = 9$
 $3 \times 4 = 12$
 $3 \times 5 = 15$
 $3 \times 6 = 18$
 $3 \times 7 = 21$
 $3 \times 8 = 24$
 $3 \times 9 = 27$
 $3 \times 10 = 30$

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 5

Exiting the program. Goodbye!

Exiting the program. Goodbye!

```
(kali㉿kali)-[~]
```

```
$ ./math_tables
```

Enter two positive integers m and n (where m <= n):

Enter m: 5

Enter n: 5

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 2

Addition table for 5:

5 + 1 = 6
5 + 2 = 7
5 + 3 = 8
5 + 4 = 9
5 + 5 = 10
5 + 6 = 11
5 + 7 = 12
5 + 8 = 13
5 + 9 = 14
5 + 10 = 15

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 5

Exiting the program. Goodbye!

```
(kali㉿kali)-[~]
```

```
$ ./math_tables
```

Enter two positive integers m and n (where m <= n):

Enter m: 4

Enter n: 4

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 3

Subtraction table for 4:

Subtraction table for 4:

4 - 1 = 3
4 - 2 = 2
4 - 3 = 1
4 - 4 = 0
4 - 5 = -1
4 - 6 = -2
4 - 7 = -3
4 - 8 = -4
4 - 9 = -5
4 - 10 = -6

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 5

Exiting the program. Goodbye!

└─(kali㉿kali)-[~]

└─\$./math_tables

Enter two positive integers m and n (where m <= n):

Enter m: 7

Enter n: 8

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 4

Division table for 7:

7 / 1 = 7.00
7 / 2 = 3.50
7 / 3 = 2.33
7 / 4 = 1.75
7 / 5 = 1.40
7 / 6 = 1.17
7 / 7 = 1.00
7 / 8 = 0.88
7 / 9 = 0.78
7 / 10 = 0.70

Division table for 8:

8 / 1 = 8.00
8 / 2 = 4.00
8 / 3 = 2.67
8 / 4 = 2.00

```
8 / 1 = 8.00
8 / 2 = 4.00
8 / 3 = 2.67
8 / 4 = 2.00
8 / 5 = 1.60
8 / 6 = 1.33
8 / 7 = 1.14
8 / 8 = 1.00
8 / 9 = 0.89
8 / 10 = 0.80
```

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 5

Exiting the program. Goodbye!

```
(kali㉿kali)-[~]
```

```
$ ./math_tables
```

Enter two positive integers m and n (where m <= n):

Enter m: 2

Enter n: 4

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 7

Error: Invalid menu choice. Please try again.

--- Mathematical Tables Menu ---

1. Print the first 10 multiples
2. Print the addition table (addends 1 to 10)
3. Print the subtraction table (subtrahends 1 to 10)
4. Print the division table (divisors 1 to 10)
5. Exit

Enter your choice (1-5): 5

Exiting the program. Goodbye!

```
(kali㉿kali)-[~]
```

```
$ ./math_tables
```

Enter two positive integers m and n (where m <= n):

Enter m: 5

Enter n: 2

Error: Both numbers must be positive integers and m must be less than or equal to n.

Step 5: - Standard Test Cable

TEST TABLE	INPUT (m, n)	MENU CHOICE	EXPECTED OUTPUT OR BEHAVIOR	RESULT (PASS/ FAIL)
1	$m=2$ $n=3$	1 (Multiples)	Prints $2 \times 1 = 2$ to $2 \times 10 = 20$, then $3 \times 1 = 3$ to $3 \times 10 = 30$	PASS
2	$m=5$ $n=5$	2 (Addition)	Prints $5 + 1 = 6$ through $5 + 10 = 15$ only	PASS
3	$m=4$ $n=4$	3 (Subtraction)	Prints $4 - 1 = 3$, $4 - 2 = 2$, down to $4 - 10 = -6$	PASS
4	$m=7$ $n=8$	4 (Division)	Prints $7 / 1 = 7.00 \dots$ $7 / 10 = 0.70$, then $8 / 1 \dots$	PASS
5	$m=2$ $n=4$	7 (Invalid)	"Error: Invalid menu choice. Please try again."	PASS
6	$m=5$ $n=2$	N/A	"Error: Both numbers must be positive... $m \leq n$ "	PASS

Step 6: - Evaluation of the algorithm

Since the inner loop iterates a constant 10 times for every integer in the sequence, the total operations scale linearly based exclusively on the size of the interval between m and n .